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Software Requirement Specification

HeritageBD – Digital Heritage Tourism & Marketplace Platform of Bangladesh

Course Code: CSE 416

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to formally describe the functional and non-functional requirements of HeritageBD, a digital heritage tourism and marketplace platform for Bangladesh. This document describes the full-stack implementation of the system, consisting of a React single-page application, a PHP REST API and a MySQL relational database. It serves as a reference for developers, instructors, evaluators and other stakeholders by defining the behaviour, constraints and operational requirements of the system.

HeritageBD provides a centralized, web-based ecosystem that connects tourists, local vendors, tour guides and platform administrators around the discovery of heritage places, traditional food, handicrafts and GI products, the booking of verified tour guides, trip planning, an interactive map, and a marketplace with a flagship “Ready When You Arrive” pre-order and pickup module.

1.2 Document Conventions

The following conventions are used throughout this document. The entire document text is justified.

- Main Titles: Font – Times New Roman, Style – Bold, Size – 14.
- Section and Subsection Headings: Font – Times New Roman, Style – Bold, Size – 12.
- Body Text: Font – Times New Roman, Size – 12.
- Functional requirements are identified as FR-nn in Section 5.7 for traceability.

1.3 Scope of Development Project

HeritageBD transforms the fragmented, manual experience of discovering and engaging with Bangladesh’s cultural heritage into a single, modern digital ecosystem. Rather than being a simple e-commerce site, it combines heritage discovery, an interactive map, tour-guide discovery and booking, trip planning, a marketplace for food, handicrafts and GI products, in-app chat, notifications and order tracking into one product journey: Sign Up → Discover → Explore Map → Find Guide → Plan Trip → Visit → Experience → Buy Local Products → Review → Share.

A central feature is “Ready When You Arrive”, which lets a tourist pre-order eligible local food, handicrafts and GI products near a destination before travelling. The tourist pays a 30% advance at booking, the vendor prepares the order, and the tourist pays the remaining 70% at pickup after QR-code verification. The advance percentage, cancellation window, refund percentage and preparation cut-off are administrator-configurable.

The system is intended for use by Tourists / Customers, Local Vendors / Sellers, Tour Guides and Administrators.

The frontend is built with React 19, Vite, TypeScript, Tailwind CSS 4, React Router, Leaflet / OpenStreetMap and Recharts. The backend is a PHP 8 REST API using PDO prepared statements and bearer-token authentication, backed by a MySQL 8 database of 46 tables with demo seed data. Future enhancements include AI-based recommendations, production

payment-gateway integration, SMS / e-mail OTP delivery, delivery tracking and a native mobile application.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
HeritageBD	The Digital Heritage Tourism & Marketplace Platform of Bangladesh
GI Product	Geographical Indication product – a good tied to a specific region of Bangladesh
RTP	"Ready To Pick" – the pre-order and pickup module
TrP	Tour planner/Trip planning
QR	Quick Response code, used to verify pickup identity
RBAC	Role-Based Access Control
API	Application Programming Interface
REST	Representational State Transfer
SPA	Single-Page Application
PDO	PHP Data Objects – the database access layer used by the backend
OTP	One-Time Password used for password reset
CORS	Cross-Origin Resource Sharing
JSON	JavaScript Object Notation – the API data format
SRS	Software Requirements Specification
ER Diagram	Entity Relationship Diagram
UI/UX	User Interface / User Experience

1.5 References

Books and Standards

- Karl E. Wiegers, Software Requirements, Microsoft Press
- Roger S. Pressman, Software Engineering: A Practitioner's Approach
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications

Web Resources

- https://react.dev
- https://vite.dev
- https://tailwindcss.com
- https://leafletjs.com
- https://www.openstreetmap.org
- https://www.php.net
- https://dev.mysql.com/doc

2. Overall Description

2.1 Product Perspective

HeritageBD is a new, standalone web application that replaces separate travel blogs, marketplaces, listing sites and word-of-mouth guide arrangements with a single connected platform. The system follows a three-tier architecture:

- **Presentation Layer:** a responsive React (Vite + TypeScript + Tailwind CSS) single-page application. A typed API client (services/api.ts) and a global AppContext manage authentication, cart and notifications.
- **Application Layer:** a PHP REST API. A front controller (index.php) matches each request against a route table and dispatches it to one controller per resource. Helper modules provide bearer-token authentication, role guards, JSON responses and in-app notifications. Responses always use the form { success, data } or { success, message }.
- **Data Layer:** a MySQL 8 / MariaDB 10.4+ relational database (utf8mb4, InnoDB) of 46 tables with foreign keys and full-text indexes.

All users – Tourists, Vendors, Tour Guides and Administrators – interact with the same centralized data source under role-based access permissions.

The use case diagram summarizes the primary interactions of each actor. Tourists discover heritage content, use the interactive map, find and book tour guides, plan trips, buy products, place “Ready To Pick” pre-orders (which include the 30% advance payment and QR-based pickup), track orders, chat and leave reviews. Vendors manage their shop and products, receive and fulfil orders, prepare pre-orders, and scan pickup QR codes to collect the remaining 70%. Tour guides manage their profile, toggle availability and location sharing, and accept or reject bookings. Administrators manage users, verify vendors and guides, manage heritage content, configure pre-order payment and refund rules, and view statistics.



Figure 1: Use Case Diagram of HeritageBD

2.2 Product Functions

The entity relationship diagram of the implemented database is shown in Figure 2 (next page). For readability it shows the principal tables; supporting tables (images, tags, languages, time slots, etc.) are listed in Section 2.7.

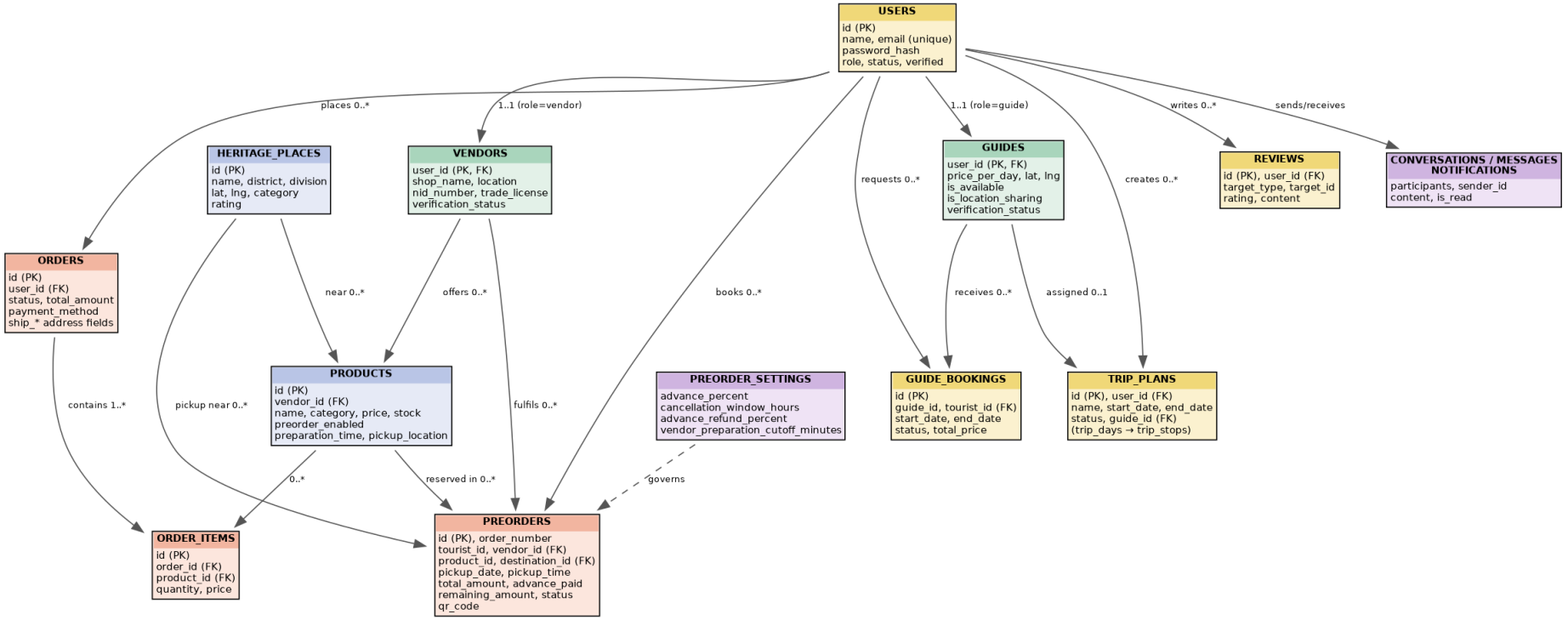


Figure 2: Entity Relationship (ER) Diagram of HeritageBD

The primary functions of HeritageBD include:

- User registration, login and role-based authorization for Tourists, Vendors, Tour Guides and Administrators, with account verification for vendors and guides
- Heritage place, food, product, event and story discovery with search and filters
- An interactive Leaflet/OpenStreetMap map showing heritage places and tour guides, extensible to vendors, products and events
- Tour guide discovery, profile viewing, booking requests and in-app chat
- Trip planning with multi-day itineraries (days and ordered stops)
- Marketplace browsing, server-side cart, checkout and delivery-based order tracking
- "Ready To Pick" pre-ordering: destination-based product discovery, 30% advance payment, vendor preparation, QR-based pickup and 70% remaining payment
- Reviews and ratings for places, products and guides, with automatic aggregate-rating updates
- In-app notifications for orders, bookings, reviews and system events
- Administrative monitoring, vendor and guide verification, content management, pre-order payment/refund configuration and statistics

The system reduces the friction of planning and experiencing heritage tourism in Bangladesh while improving transparency and trust between tourists, vendors and guides.

2.3 User Classes and Characteristics

Tourist / Customer

- Discovers heritage places, food, handicrafts, GI products, events and stories; saves favourite places
- Uses the interactive map to find nearby attractions and available guides
- Finds, evaluates and books verified tour guides, and chats with them
- Plans multi-day trips; buys products for delivery or pre-orders eligible products for pickup
- Tracks orders and pre-orders; writes reviews and ratings

Local Vendor / Seller

- Registers a shop (shop name, location, NID and trade-licence details) and awaits admin verification
- Adds products, images, prices and stock; marks eligible products for "Ready When You Arrive" with preparation time, pickup location and pickup time slots
- Receives and fulfils standard orders and pre-orders; updates their status
- Scans a tourist's pickup QR code to verify the order and collect the remaining 70%

Tour Guide

- Registers with NID and guide-licence details and awaits admin verification
- Specifies languages, specializations, certifications, cover areas, tour types and price per day

- Explicitly toggles “Available for Tour” and location sharing
- Accepts or rejects booking requests and chats with tourists
- Guide also plan tour for tourists.

Administrator

- Accounts are created by the platform (self-registration as admin is blocked)
- Manages users, vendors and guides, including verification (approve / reject) and suspension
- Manages heritage places, events, stories, products and reviews
- Configures pre-order settings: advance percentage, payment methods, cancellation window, refund percentage, preparation cut-off, maximum pickup slots and cash-at-pickup
- Monitors orders and pre-orders and views platform statistics and reports

2.4 Operating Environment

The system is a web application supporting modern browsers (Google Chrome, Mozilla Firefox, Safari, Microsoft Edge) on desktop, tablet and mobile devices.

Hardware Requirements (Client)

- Processor: Dual-core or higher
- RAM: Minimum 4 GB
- Storage: Browser cache only; no local installation required

Software Requirements

- Client OS: Windows / macOS / Linux / iOS / Android with a supported browser
- Frontend: React 19, Vite, TypeScript, Tailwind CSS 4, React Router 7, Leaflet / react-leaflet, Recharts, Lucide React
- Backend: PHP 8.0+ with the pdo_mysql extension; Apache (mod_rewrite), Nginx, or the PHP built-in server for development
- Database: MySQL 8.0+ or MariaDB 10.4+
- Internet connectivity is required

2.5 Assumptions and Dependencies

Assumptions

- Users possess basic smartphone or computer literacy
- Stable internet connectivity is available to tourists, vendors and guides
- Data entered by vendors, guides and tourists is accurate
- The system operates continuously (24/7)
- Mobile money (bKash, Nagad) and card payments are available to end users

Dependencies

- Web hosting for the frontend, PHP API and MySQL database
- OpenStreetMap tile services for the interactive map

- Secure HTTPS configuration
- Third-party payment gateway integration (bKash, Nagad, card processors) for production
- An e-mail / SMS provider for OTP delivery in production

2.6 Requirement

Software Configuration

Component	Technology
Frontend	React 19, Vite 8, TypeScript 5, Tailwind CSS 4, React Router 7, Leaflet, Recharts, Lucide React
Backend	PHP 8.0+ (REST API, PDO, bcrypt password hashing)
Database	MySQL 8.0+ / MariaDB 10.4+ (utf8mb4, InnoDB)
Web Server	Apache with mod_rewrite (.htaccess provided), Nginx, or PHP built-in server
Configuration	VITE_API_URL (frontend); DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASS, CORS_ORIGIN (backend)

Hardware Configuration (Server)

- Processor: Dual-core CPU or higher
- RAM: 4 GB or more
- Storage: 40 GB or higher

2.7 Data Requirement

The system processes input data such as user credentials, vendor and guide verification details, heritage place and product content, product selections, cart contents, shipping addresses, pre-order pickup details (destination, date, time, quantity), payment method choices, bookings, trips, reviews and chat messages. Outputs include heritage and product listings, map data, guide availability, order and pre-order status, QR pickup codes, notifications, and statistics. All data is stored in a MySQL relational database of 46 tables grouped as follows.

Group	Tables	Table names
Users & roles	12	users, auth_tokens, password_resets, tourist_saved_places, vendors, guides, guide_languages, guide_specializations, guide_certifications, guide_cover_areas, guide_tour_types, admins
Heritage places	4	heritage_places, place_images, place_tags, place_nearby
Marketplace	9	products, product_images, product_tags, product_specifications, product_time_slots, product_near_destinations, cart_items, orders, order_items
Pre-orders	2	preorder_settings, preorders
Guide bookings	2	guide_bookings, guide_booking_destinations
Events	3	events, event_images, event_tags
Stories	2	stories, story_tags

Group	Tables	Table names
Reviews	2	reviews, review_photos
Trip planner	4	trip_plans, trip_destinations, trip_days, trip_stops
Notifications	1	notifications
Chat	4	conversations, conversation_participants, messages, message_attachments
Categories	1	categories

2.8 SWOT Analysis

The SWOT analysis identifies the internal strengths and weaknesses of HeritageBD, and the external opportunities and threats that may affect it.

Strengths

HeritageBD centralizes heritage discovery, tour-guide booking, trip planning and a local marketplace – including the "Ready To Pick" pre-order system – in one platform. Admin verification of vendors and guides and a rating system build trust. The 30/70 payment split protects vendors (advance commitment) and tourists (majority paid after verified pickup). The map ties places and guides together spatially. A clean REST API and relational schema make the system maintainable and extensible.

Weaknesses

The platform depends on stable internet connectivity for tourists, vendors and guides, and pickup verification depends on the tourist being able to display a QR code. New vendors and guides may need onboarding support. As a centralized system, a server or database outage affects bookings and pre-orders nationwide. Payments, OTP delivery and QR scanning currently use simulated or simplified flows (see Section 6.5).

Opportunities

Integration of bKash, Nagad and card gateways, SMS / e-mail OTP, AI-based recommendations, delivery tracking for marketplace orders, a native mobile application and cloud deployment can grow the platform across more districts of Bangladesh.

Threats

Cybersecurity threats such as unauthorized access, payment fraud or data breaches could compromise user and payment data, particularly around advance-payment and QR flows. Outages could disrupt guide availability and pre-order updates. Lack of backups could lose order and review history. Competing platforms and changes in Bangladesh's mobile payment ecosystem may require continuous adaptation.

3. External Interface Requirements

3.1 Graphical User Interface (GUI)

HeritageBD provides a consistent, mobile-first graphical interface across all roles using a shared design system (navbar, footer, buttons, cards, badges, modals, rating stars, tables, charts, forms). Key interfaces, with their routes, are:

Interface	Route / description
Landing page	/ – hero search, featured places, guides, products, events and stories
Explore & Places	/explore, /places, /places/:id – search, category filters, place detail with gallery, reviews, nearby guides and pre-order products
Interactive map	/map – Leaflet map with toggleable Heritage Places and Tour Guides layers and marker detail cards
Tour guides	/guides, /guides/:id – listing, profile, reviews and booking panel; /guide-dashboard and /guide/tour-planner for guides
Marketplace	/shop, /shop/:id, /cart, /checkout, /order-tracking
Pre-order flow	/preorder/checkout, /preorder/tracking/:orderId, /preorder/qr/:orderId
Events & Stories	/events, /stories, /stories/:id
Trip planner	/planner – day-by-day itinerary building
Chat	/chat – conversation list and messages
Authentication	/login, /signup, /forgot-password, /onboarding/vendor, /onboarding/guide, verification-status, /suspended, /session-expired
Profile & settings	/profile, /settings
Vendor dashboard	/vendor – products, orders, Pre-Orders tab and QR scanner
Admin dashboard	/admin, /admin/guides – sidebar sections for dashboard, users, vendors, guides, verification center, places, products, orders, pre-orders, reviews, events, stories, reports, pre-order settings and settings

All screens are responsive across desktop, tablet and mobile.

3.2 Software Interfaces

- Frontend ↔ Backend: JSON over HTTP(S) through the REST API under the /api prefix. The base URL is set by VITE_API_URL.
- Backend ↔ Database: PDO with prepared statements over the pdo_mysql driver; connection details come from environment variables.
- Frontend ↔ Map tiles: OpenStreetMap tile servers via Leaflet.

3.3 Communication Interfaces

- Authenticated requests carry the header Authorization: Bearer <token>. Tokens are 256-bit random values stored in auth_tokens with a 30-day expiry.
- CORS headers are emitted for the configured origin (CORS_ORIGIN, default *); preflight OPTIONS requests return 204.

- HTTP status codes used: 200, 201, 204, 401 (unauthenticated), 403 (forbidden), 404, 409 (duplicate e-mail), 422 (validation) and 500.

4. System Features

This section describes the major capabilities of HeritageBD: the interactive map and guide discovery, the “Ready When You Arrive” pre-order module, the trip planner, order tracking, account verification, guide booking, and reviews, notifications and chat.

4.1 Interactive Map & Tour Guide Discovery

HeritageBD includes a full-height interactive map (Leaflet / OpenStreetMap) with a sidebar of layer filters. Markers represent heritage places and available tour guides; the design extends to food, vendors, products and events. Clicking a marker opens a card with name, rating, image and description and calls to action. Guides should appear on the map only when they have enabled “Available for Tour” and location sharing.

Features

- Togglable map layers with item counts
- Marker detail cards with rating and links to profile or detail pages
- Guide availability (is_available) and location sharing (is_location_sharing) controlled by the guide
- Search and filter of guides by location, availability and name

Benefits

- Gives tourists a single spatial view of heritage places and guides
- Builds trust through verified guides and visible ratings
- Protects guide privacy through opt-in location sharing

4.2 “Ready To Pick” -- Pre-Order & Pickup System

“Ready When You Arrive” allows a tourist to pre-order eligible local food, handicrafts and GI products near a heritage destination so the vendor can prepare in advance: “Order before you go. Pick up when you arrive.” At every stage the tourist sees what was ordered, where and when to pick it up, how much is paid and how much remains.

4.2.1 Payment Model

Every pre-order requires an advance payment (default 30%) at booking and the remaining balance (default 70%) at pickup. For a product total of ৳1,000, the advance is ৳300 and the remaining balance is ৳700. The split is stored per pre-order (total_amount, advance_percent, advance_paid, remaining_amount) and displayed at every stage. The advance may be paid by bKash, Nagad or card; the remainder at pickup by bKash, Nagad, card or cash.

4.2.2 Tourist Journey

Discover Destination → View Nearby Pre-order Products → Select Product → Select Pickup Date, Time and Quantity → View Advance → Pay Advance → Order Confirmed with QR → Vendor Prepares → Ready for Pickup → Tourist Arrives → Show QR → Vendor Verifies → Pay Remaining → Receive Product → Completed → Rate & Review.

4.2.3 Vendor Journey

The vendor receives a new pre-order notification, accepts it, prepares the item within the stated preparation time and marks it ready. When the tourist arrives, the vendor scans the pickup QR code, the system shows the verified order and remaining amount, and the vendor collects the remaining payment before handover.

4.2.4 Order Status Timeline

Pre-orders use a pickup-oriented timeline distinct from delivery orders. The stored status values are:

No.	Status	Meaning
1	confirmed	Pre-order created
2	advance_paid	Advance payment received
3	vendor_accepted	Vendor accepted the pre-order
4	preparing	Vendor is preparing the item
5	ready_for_pickup	Item is ready
6	arrived	Tourist arrived at the pickup location
7	qr_verified	Vendor verified the pickup QR code
8	pickup_payment_pending	Remaining payment awaited
9	picked_up	Item handed over
10	completed	Order completed
–	cancelled / refunded	Cancellation requested or refund issued

4.2.5 Integration with Product Pages, Map and Trip Planner

Eligible products (preorder_enabled) show a pre-order badge with preparation time, pickup location and available time slots. Products are linked to nearby heritage destinations (product_near_destinations), and destination pages list pre-order products. Pre-orders and trip plans are both attached to the tourist so confirmed pickups can be shown alongside the itinerary.

4.2.6 Cancellation and Refunds

A tourist may request cancellation before the vendor begins preparation. The platform does not hard-code a refund policy: the administrator configures the cancellation window (hours), advance refund percentage and vendor preparation cut-off (minutes) in preorder_settings, and the UI reflects the current policy. Seed defaults are 2 hours, 80% and 30 minutes.

Features

- Destination-based discovery of pre-order products with pickup location, preparation time, price and availability
- Two purchasing modes on eligible products: Buy Now / Delivery and Pre-order / Pickup
- Multi-step pre-order checkout (pickup details, summary, advance payment, confirmation)

- Unique order number (HB + date + random code) and order-specific QR code per pre-order
- Tourist QR screen and vendor QR scanner screen; remaining-payment capture at pickup
- Dedicated pre-order tracking timeline
- Administrator settings for advance percentage, payment methods, cancellation, refund, cut-off, pickup slots and cash at pickup

Benefits

- Reduces waiting time for tourists and wasted preparation for vendors
- Keeps payment expectations transparent
- Increases vendor commitment through an upfront advance while protecting tourists
- Creates a discovery path from heritage destinations to local vendors

4.3 Trip Planner

Tourists create named trips (for example “Dhaka Heritage Weekend”) with a start and end date, a status (draft, planned, ongoing, completed), optional assigned guide, destinations, and a list of days each containing ordered stops (place, duration, notes). Trips can be listed, viewed, updated and deleted.

Features

- Day-by-day itinerary with ordered stops (sort_order)
- Stops linked to heritage places
- Optional guide assignment
- Trip status lifecycle

Benefits

- Gives tourists a single, coherent view of their visit
- Reduces planning effort

4.4 Order Tracking System

Standard marketplace orders are created from the server-side cart at checkout. Stock is validated for every line and decremented inside a database transaction; the cart is then cleared and the tourist notified. Delivery orders progress through: placed → confirmed → preparing → shipped → out_for_delivery → delivered (or cancelled). A tracking number and an estimated delivery date (five days after placement) are stored. Tourists see their own orders, vendors see orders containing their products, and administrators see all orders. Pre-orders use the separate timeline in Section 4.2.4.

Features

- Order number, items, shipping address, payment method (bKash, Nagad, card, cash on delivery), status and tracking number
- Vendor / admin status updates with automatic tourist notification
- Order history

Benefits

- Improves transparency and reduces communication delays
- Enhances customer trust in the platform

4.5 Authentication, Roles & Account Verification

Users register as Tourist, Vendor or Guide; administrator accounts cannot self-register. Tourists are active immediately. Vendors and guides are created with status “pending” and a pending verification record containing NID and trade-licence / guide-licence details, and must be approved by an administrator (verified) or rejected (account suspended). Login is refused for suspended accounts. A successful login or registration issues a 30-day bearer token; logout revokes it. Password reset uses a six-digit OTP valid for 15 minutes.

4.6 Tour Guide Booking

A tourist sends a booking request to a guide with tour type, start and end dates, group size, destinations and a message. The booking starts as pending; the guide accepts or rejects it and can later mark it completed; either party can cancel. Each status change notifies the other party. Tourists see their bookings and guides see bookings addressed to them.

4.7 Reviews, Notifications & Chat

- Reviews: authenticated users submit a 1–5 star rating, optional title, text and photos for a place, product or guide. The target’s average rating and review count are recalculated on every new review.
- Notifications: in-app notifications (types booking, order, review, system, message) are created for events such as new pre-orders, order and booking updates and verification outcomes; they can be marked read individually or all at once.
- Chat: users start conversations and exchange messages; unread counts are tracked per participant.

4.8 Administration

The administrator dashboard provides counters (users by role, pending guides and vendors, places, products, orders, bookings, pre-orders, monthly revenue), user listing with role and status filters, vendor and guide verification, management of places, products, events, stories and reviews, pre-order monitoring, reports with charts, and the pre-order settings form.

5. Other Non-Functional Requirements

5.1 Performance Requirements

- Fast response under normal load, including map rendering and search/filter results
- List endpoints are paginated (limit capped at 200) and use database indexes and FULLTEXT indexes for search
- Support large volumes of heritage content, product listings and concurrent bookings during peak seasons and festivals
- Minimal downtime, particularly around active pre-order pickup windows

5.2 Safety Requirements

- Regular database backups covering users, orders, pre-orders and payment records
- Multi-table writes (registration, checkout) run inside database transactions with rollback on failure
- Protection against server power failure using UPS / redundant hosting

5.3 Security Requirements

- Passwords are stored only as bcrypt hashes (password_hash / password_verify)
- Authentication by random 256-bit bearer tokens with 30-day expiry, revoked on logout
- Role-based guards (Tourist, Vendor, Guide, Admin) on protected endpoints; suspended users cannot log in
- All SQL uses PDO prepared statements to prevent SQL injection
- CORS origin restricted to the production frontend; all traffic served over HTTPS in production
- Explicit, revocable permission for guide location sharing
- QR pickup codes are unique per pre-order and must be single-use and validated server-side
- Prevention of unauthorized modification of pre-order status and payment records

5.4 Requirement Attributes

- Component-based, reusable frontend architecture with a typed API client
- Clean separation of frontend, REST API and database
- High maintainability and usability across desktop, tablet and mobile
- Bangla-capable storage (utf8mb4)

5.5 Business Rules

- Only administrators can verify or reject vendor and guide accounts
- Administrator accounts cannot self-register
- Tourists can pre-order only products marked pre-order-enabled by the vendor
- A pre-order requires an advance payment (default 30%); the remainder is due at verified pickup
- Vendors maintain accurate product, price and stock records, including preparation time for pre-order items
- Guides must explicitly enable availability and location sharing before being shown on the map
- Cancellation window, advance refund percentage and preparation cut-off are configured by the administrator, not hard-coded
- Checkout fails if any item’s stock is insufficient
- Only one account may exist per e-mail address

5.6 User Requirements

Users are expected to have basic smartphone or computer and internet knowledge. Vendors and guides should understand how to manage their profile, availability, inventory and pricing. Administrators need deeper understanding to manage verification, pre-order settings, backups and monitoring.

5.7 Functional Requirements

Functional requirements describe the specific services HeritageBD must perform.

User Management Requirements

FR-01 User Registration: The system shall allow tourists, vendors and guides to register with name, e-mail, password, phone and role; vendor and guide registrations shall also capture shop / guide details and NID and licence numbers.

FR-02 User Authentication: The system shall verify credentials at login, issue a bearer token, reject suspended accounts, and allow logout to revoke the token.

FR-03 Role-Based Access Control: The system shall enforce permissions by role (tourist, vendor, guide, admin) on every protected endpoint.

FR-04 Vendor and Guide Verification: The administrator shall approve or reject pending vendors and guides; approval marks the account verified and active, rejection suspends it, and the user is notified.

FR-05 Password Reset: The system shall generate a six-digit OTP valid for 15 minutes and allow the password to be reset with it.

FR-06 Profile Management: Users shall view and edit their name, phone and avatar, and tourists shall save and unsave heritage places.

Heritage Discovery & Content Requirements

FR-07 Browsing and Search: Users shall browse and search places, products, events and stories with filters (e.g. category, district, division, featured, price, rating, upcoming).

FR-08 Place Details: The system shall show gallery, description, historical background, significance, opening hours, entry fee, best time to visit, tags, nearby attractions, reviews, nearby guides and pre-order products.

FR-09 Content Management: Administrators shall create, update and delete heritage places, and manage events and stories.

FR-10 Categories: The system shall provide categories with item counts for the Explore page.

Interactive Map Requirements

FR-11 Map Rendering: The system shall render an interactive map showing heritage places and available tour guides using OpenStreetMap tiles.

FR-12 Layer Filtering: Users shall toggle each marker layer on or off.

FR-13 Marker Detail Cards: Clicking a marker shall open a card with details and links to the place or guide.

Tour Guide Management Requirements

FR-14 Guide Profile: Guides shall maintain bio, location, price per day, languages, specializations, certifications, cover areas and tour types.

FR-15 Availability & Location Sharing: Guides shall toggle availability and location sharing; the system shall expose a guide's coordinates only when both are enabled.

FR-16 Booking Requests: Tourists shall send booking requests; guides shall accept or reject them; status changes shall notify the other party.

FR-17 Guide Dashboard: Guides shall see bookings, upcoming tours, messages and reviews.

Trip Planner Requirements

FR-18 Trip Creation: Tourists shall create named trips with dates, destinations and status.

FR-19 Itinerary Management: The system shall store days and ordered stops per trip and allow update and deletion of trips.

Marketplace / Product Management Requirements

FR-20 Product Listing: Vendors shall add, edit and delete products with images, price, category, description, stock, tags and specifications.

FR-21 Product Browsing: Users shall browse products filtered by category, district, price range, pre-order availability and sorted by price, rating or newest.

FR-22 Cart: Authenticated users shall add, update, remove and clear cart items stored on the server.

FR-23 Checkout: The system shall create an order from the cart with a shipping address and payment method (bKash, Nagad, card, cash on delivery).

FR-24 Inventory Validation: The system shall verify stock before confirming an order and decrement it atomically.

Pre-Order ("Ready When You Arrive") & Payment Requirements

FR-25 Pre-order Eligibility: Vendors shall mark products as pre-order eligible and specify preparation time, pickup location, pickup time slots, nearby destinations and maximum advance days.

FR-26 Destination-Based Discovery: The system shall list pre-order-eligible products near a selected destination.

FR-27 Pickup Scheduling: Tourists shall select pickup date, time and quantity.

FR-28 Advance Payment Calculation: The system shall calculate and display the total, advance amount and remaining amount.

FR-29 Advance Payment Processing: The system shall record the advance payment (bKash, Nagad or card) and confirm the pre-order.

FR-30 Order Number and QR Code: The system shall generate a unique order number and an order-specific QR code for each pre-order.

FR-31 Pre-order Status Updates: Vendors shall update status through accepted, preparing and ready; the system shall notify the other party at each change.

FR-32 Pickup Verification: Vendors shall verify the tourist's QR code before collecting the remaining payment; a code shall be usable once.

FR-33 Remaining Payment: The system shall record the remaining payment (bKash, Nagad, card or cash) and mark the pre-order completed.

FR-34 Cancellation & Refunds: Tourists shall request cancellation before preparation begins; the system shall apply the administrator-configured refund policy.

FR-35 Pre-order Settings: Administrators shall configure advance percentage, payment methods, cancellation window, refund percentage, preparation cut-off, maximum pickup slots and cash-at-pickup.

Order Tracking Requirements

FR-36 Delivery Order Tracking: The system shall track orders through placed, confirmed, preparing, shipped, out for delivery and delivered, with a tracking number.

FR-37 Pre-order Tracking: The system shall track pre-orders through the timeline in Section 4.2.4.

FR-38 Order History: The system shall retain previous orders and pre-orders for each user (role-aware listing).

Review & Rating Requirements

FR-39 Review Submission: Authenticated users shall submit 1–5 star ratings, text and optional photos for places, products and guides.

FR-40 Aggregate Rating: The system shall recalculate the target's average rating and review count after each review.

FR-41 Review Display: The system shall display reviews with user, date, rating and comment.

Notification & Chat Requirements

FR-42 Notifications: The system shall create in-app notifications for orders, pre-orders, bookings, verification outcomes and messages, and allow marking as read.

FR-43 Chat: Users shall start conversations, send and read messages, and see unread counts.

Admin & Reporting Requirements

FR-44 User Management: Administrators shall list users filtered by role and status and suspend or reinstate accounts.

FR-45 Pre-order Management: Administrators shall view all pre-orders across the platform.

FR-46 Statistics: The system shall provide counters for users, vendors, guides, pending verifications, places, products, orders, bookings, pre-orders and monthly revenue.

FR-47 Reports: The system shall present charts and reports for orders, sales, destinations and bookings.

Security Requirements

FR-48 Secure Login: The system shall provide secure login and logout with token expiry and a session-expired screen.

FR-49 Unauthorized Access Prevention: The system shall return 401 or 403 for requests outside the user's role or ownership.

FR-50 Activity Logging: The system shall maintain logs of important user and payment events for auditing.

FR-51 Backup and Recovery: The system shall support regular database backup and recovery.

6. Other Requirements

6.1 Data and Category Requirements

Users are categorized as Tourist, Vendor, Tour Guide and Administrator. Content is categorized as follows:

Content	Categories
Heritage places	mosque, fort, palace, temple, monument, archaeological, natural, museum
Products	gi-product, handicraft, clothing, food, jewelry, pottery, other
Events	cultural, food, market, heritage, festival, religious
Stories	heritage, food, culture, travel, festival, artisan
Reviews target	place, product, guide
Notification types	booking, order, review, system, message

6.2 Appendix

- A: Acronyms and Abbreviations
- B: Business Rules
- C: Categories (Places, Products, Events, Stories)
- D: Data Requirements (46-table schema)
- E: REST API Reference (below)
- G: Graphical User Interface
- N: Non-Functional Requirements
- P: "Ready To Pick" Pre-Order Business Logic
- S: Security and System Features

Appendix E: REST API Reference (prefix /api)

Method	Path	Access	Description
POST	/auth/register	Public	Create tourist / vendor / guide account

Method	Path	Access	Description
POST	/auth/login	Public	Login; returns user and token
POST	/auth/logout	Auth	Revoke current token
GET / PUT	/auth/me	Auth	Get / update own profile
POST	/auth/forgot-password	Public	Request OTP
POST	/auth/reset-password	Public	Reset password with OTP
GET	/places, /places/{id}	Public	List / search / detail
POST / PUT / DELETE	/places[/{id}]	Admin	Manage places
POST / GET	/places/{id}/save, /places/saved	Auth	Toggle / list saved places
GET	/places/{id}/reviews	Public	Reviews for a place
GET	/categories	Public	Categories with counts
GET	/guides, /guides/{id}	Public	List / detail of guides
PUT	/guides/{id}	Self / Admin	Update guide profile
POST / GET / PUT	/bookings[/{id}]	Auth	Create (tourist), list, update booking status
GET	/products, /products/{id}	Public	List / filter / detail
POST / PUT / DELETE	/products[/{id}]	Vendor / Admin	Manage products
GET / POST / DELETE	/cart	Auth	View / add / clear cart
PUT / DELETE	/cart/{productId}	Auth	Update quantity / remove item
POST / GET	/orders	Auth	Checkout / list orders (role-aware)
GET / PUT	/orders/{id}	Auth / Vendor, Admin	Order detail / update status
GET / PUT	/preorders/settings	Public / Admin	Read / update pre-order settings
POST	/preorders	Tourist	Create pre-order
GET	/preorders, /preorders/{id}	Auth	List (role-aware) / detail
PUT	/preorders/{id}	Participant / Admin	Update status, payment method, cancellation, refund
GET	/events, /events/{id}	Public	Events list / detail
GET	/stories, /stories/{id}	Public	Stories list / detail
GET / POST	/reviews	Public / Auth	List by target / create review
GET / POST	/trips	Auth	List / create trips
GET / PUT / DELETE	/trips/{id}	Auth	Trip detail / update / delete
GET / PUT	/notifications, /notifications/read-all, /notifications/{id}/read	Auth	List / mark read
GET / POST	/conversations	Auth	List / start conversation
GET / POST	/conversations/{id}/messages	Auth	Read / send messages
GET	/admin/stats, /admin/users	Admin	Counters / user list
GET / PUT	/admin/guides/pending, /admin/guides/{id}/verify	Admin	Pending guides / verify or reject

Method	Path	Access	Description
GET / PUT	/admin/vendors, /admin/vendors/pending, /admin/vendors/{id}/verify	Admin	Vendors / verify or reject

6.3 Glossary

This section defines important terms used throughout this SRS.

Administrator: A privileged user who manages users, vendors, guides, content, pre-order settings and monitors the platform.

Tourist: A user who discovers heritage content, uses the map, books guides, plans trips, buys or pre-orders products, tracks orders and writes reviews.

Vendor: A local seller who manages a shop, lists products and fulfils standard orders and pre-orders.

Tour Guide: A verified individual who offers guided tours, manages availability and location sharing and responds to bookings.

Database: The MySQL relational database that stores accounts, content, orders, pre-orders, bookings, reviews, notifications and messages.

Heritage Place: A historical or cultural site with a detail page, map location, nearby content and reviews.

Product: An item listed by a vendor – GI product, handicraft, clothing, food, jewelry, pottery or other – that may support delivery, pre-order pickup or both.

Order: A request by a tourist to purchase products from the cart, with items, shipping address, payment method, total and status.

Pre-order ("Ready When You Arrive"): An order type in which a tourist reserves a product ahead of a visit, pays an advance at booking and the remainder at verified pickup.

Advance Payment: The portion of a pre-order total (default 30%) paid at booking.

Remaining Payment: The balance (default 70%) paid at verified pickup.

QR / Pickup Code: A unique order-specific code generated for each pre-order and scanned by the vendor to verify pickup.

Trip Plan: A tourist-created itinerary of days and stops, optionally with an assigned guide.

Review: A rating and written comment for a place, product or guide.

Bearer Token: A random secret sent in the Authorization header to authenticate API requests.

Interface: The point of interaction between a user and HeritageBD, including pages, dashboards and checkout flows.

Role: A set of permissions (Tourist, Vendor, Tour Guide, Administrator) enforced through RBAC.

6.4 Class Diagram

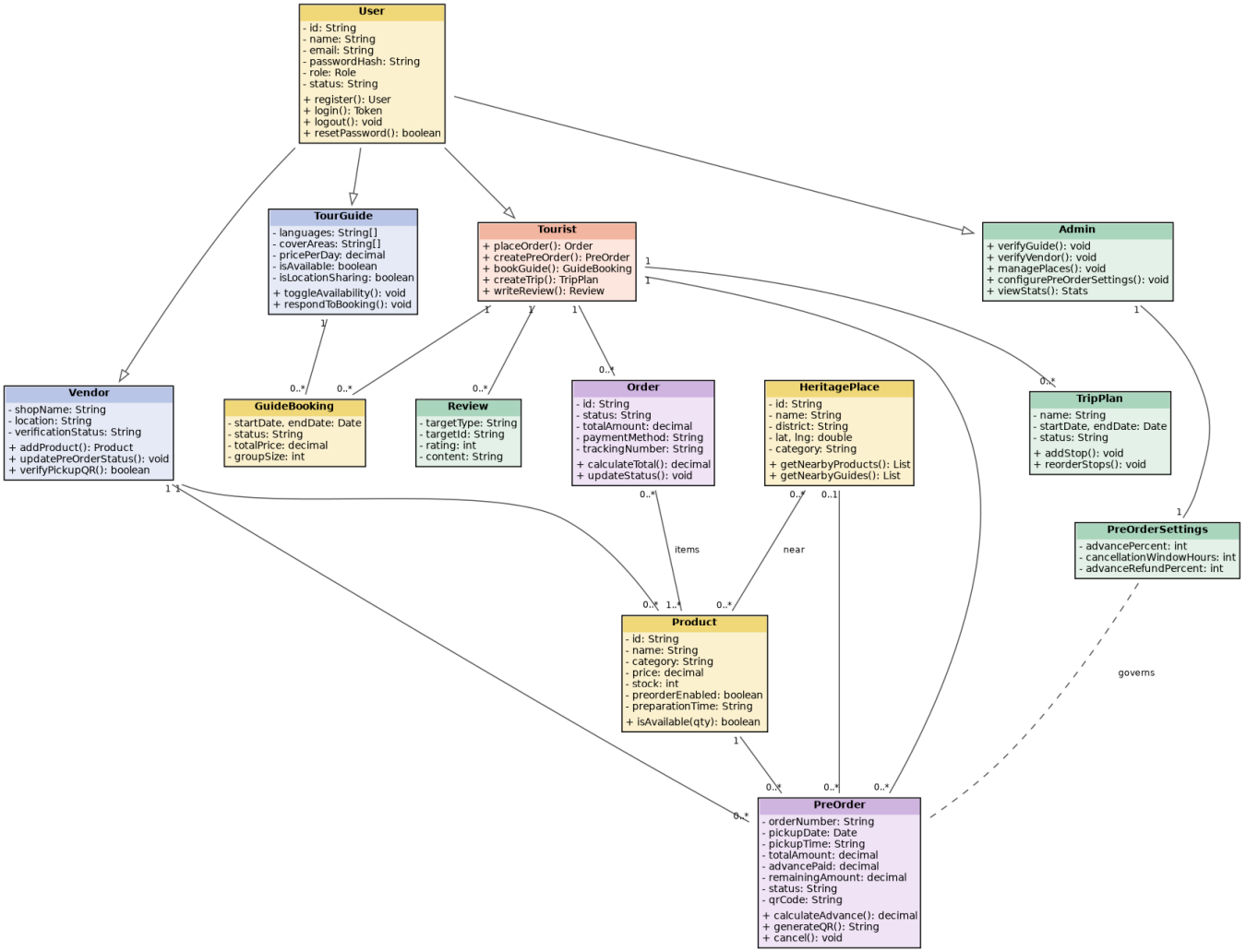


Figure 3: Class Diagram of HeritageBD

6.5 Implementation Status and Known Limitations

This section records how the current build relates to the requirements above so that evaluators and future developers have an accurate baseline. All 46 tables, the REST API (67 routes) and demo seed data are implemented; the items below are simplified or not yet complete.

Area	Current state / planned improvement
Payments	bKash, Nagad, card and cash are recorded as payment methods; no live gateway is integrated. A pre-order is stored as advance_paid when created. Production gateway integration is planned.
Advance percentage	The pre-order endpoint uses the advance percentage sent by the client (default 30). It should read advance_percent from preorder_settings so the administrator setting is authoritative.
QR verification	A QR value is generated per pre-order and shown to the tourist; the vendor dashboard provides the scanner interface. A server-side endpoint that validates the code and enforces single use is not yet implemented.
Status transitions	Pre-order and order status updates are accepted from authorized participants without validating the allowed sequence or vendor ownership. Transition rules should be enforced on the server.
Cancellation & refunds	Cancellation reason and refund amount can be stored; automatic calculation from the configured window and refund percentage is not yet performed on the server.
OTP delivery	The reset OTP is returned in the API response (demo mode). E-mail / SMS delivery is planned.
Guide location privacy	Guide coordinates are currently returned by the guide endpoints regardless of the location-sharing flag. The API should omit coordinates unless both availability and location sharing are enabled.
Map layers	The map renders Heritage Places and Tour Guides. Layers for food, vendors, products, events and a pre-order filter are specified but not yet implemented.
Screens using local state	The multi-step marketplace checkout, guide booking panel, guide dashboard toggles and parts of the admin dashboard summary figures are not yet fully connected to their API endpoints; the endpoints exist.
Reviews	Review targets are places, products and guides. Vendor reviews, vendor/guide replies and restriction of reviews to verified purchasers are future work.
Activity logging & export	An audit log table and PDF export of reports and invoices are not yet implemented.
Token handling	Tokens last 30 days without rotation; CORS defaults to * until CORS_ORIGIN is set. Both should be tightened for production.